

Experiment No. (2):

Write a JAVA Program to create an abstract class named Shape that contains two integers and an empty method named print Area(). Provide three classes named Rectangle, Triangle, and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method print Area() that prints the area of the given Shape.

AreaRTC.java

```
abstract class Shape
```

```
{  
    public int x, y;  
    public abstract void printArea();  
}
```

```
class Rectangle extends Shape
```

```
{  
    public void printArea()  
    {  
        System.out.println("Area of Rectangle is: " + x * y);  
    }  
}
```

```
class Triangle extends Shape
```

```
{  
    public void printArea()  
    {  
        System.out.println("Area of Triangle is: " + (x * y) / 2);  
    }  
}
```

```
class Circle extends Shape
```

```
{  
    public void printArea()  
    {
```

```
        System.out.println("Area of Circle is " + (22 * x * x) / 7);
    }
}
```

```
public class AreaRTC
{
    public static void main(String[] args)
    {
        Rectangle r = new Rectangle();
        r.x = 5;
        r.y = 4;
        r.printArea();
        System.out.println("");

        Triangle t = new Triangle();
        t.x = 3;
        t.y = 4;
        t.printArea();
        System.out.println("");

        Circle c = new Circle();
        c.x = 2;
        c.printArea();
        System.out.println("");
    }
}
```

Output:

Area of Rectangle is: 20

Area of Triangle is: 6

Area of Circle is 12

Viva Questions:

1. What is Abstraction?

Abstraction is a process of hiding the implementation details and showing only functionality to the user.

Another way, it shows only important things to the user and hides the internal details for example sending sms, you just type the text and send the message. You don't know the internal processing about the message delivery.

Abstraction lets you focus on what the object does instead of how it does it.

Ways to achieve Abstraction

There are two ways to achieve abstraction in java

- Abstract class (0 to 100%)
- Interface (100%)

2. What is the Abstract class ?

A class that is declared with abstract keyword, is known as abstract class in java. It can have abstract and non-abstract methods (method with body).

- It needs to be extended and its method implemented.
- It cannot be instantiated.

Abstract Method

:A method that is declared as abstract and does not have implementation is known as abstract method.

abstract void printStatus();//no body and abstract

A **factory method** is the method that returns the instance of the class.

Note: If you are extending any abstract class that have abstract method, you must **either provide** the **implementation of the method** or make this **class abstract**.